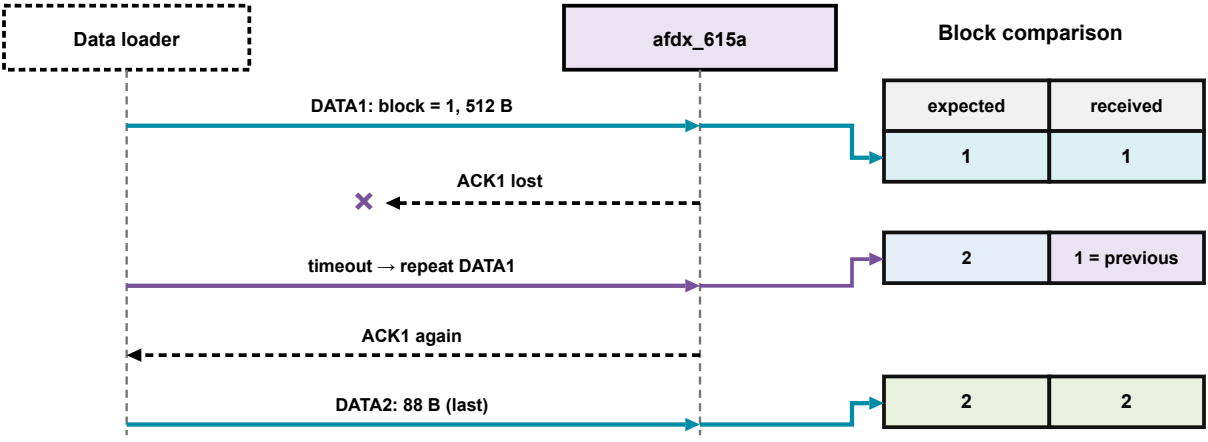
(a) Lost ACK1: resend the block, never append it twice



(b) DATA_DISPATCH selects copying or acknowledgment only

Arrival	FSM action	RAM write	loaded	CRC (hex)
DATA1	DATA_COPY	[0:511] ← 512 B	0 → 512	56EE
repeat DATA1	ACK → WAIT_PACKET	none	512 → 512	56EE
DATA2	COPY → file check	[512:599] ← 88 B	512 → 600	B166

DATA1 progress status completes before ACK1; final length / CRC checks precede ACK2.